

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi – 590 018



ASSIGNMENT REPORT

ON

“SMART FLOW AI ASSISTANT FOR STUDY”

Submitted in the view of Academics fulfillment for 1st Year assignment submission

SUBMITTED BY:

<i>Name of the Student</i>	<i>USN</i>
VISHNU	4UB25EC120

Subject:

INTRODUCTION TO AI AND APPLICATIONS

(1BAIA203)

G - Section



Department of Studies in Electronics & Communication

Engineering

2025-26

“SMART FLOW AI ASSISTANT FOR STUDY”

Abstract:

SmartFlow AI Assistant for Study is an AI-powered educational assistant developed using **n8n workflow automation, OpenAI ChatGPT**, and memory management. The system helps students by answering academic queries, providing study-related information, and maintaining conversational context through memory. The assistant acts as a virtual study companion that enables users to obtain quick and accurate responses to educational questions. The project demonstrates the integration of artificial intelligence with workflow automation to create an efficient and user-friendly learning assistant.

Introduction:

Artificial Intelligence has transformed the way students learn and access information. Traditional search methods often require significant time and effort to find relevant answers. **SmartFlow AI Assistant for Study** is designed to simplify this process by providing instant responses to user queries. The system utilizes OpenAI's language model integrated with n8n workflow automation to deliver intelligent and contextual responses. The assistant can remember previous interactions and provide personalized support to users.

Problem Statement:

Students frequently face difficulties in finding accurate study materials, clarifying concepts, and obtaining instant academic support. Existing methods require searching through multiple websites and resources, which can be time-consuming. There is a need for an intelligent assistant that can provide quick, reliable, and conversational academic assistance.

Objectives:

- ✓ To develop an AI-powered study assistant.
- ✓ To provide instant responses to educational queries.
- ✓ To maintain conversational memory for better user experience.
- ✓ To automate query processing using workflow automation.
- ✓ To enhance student learning efficiency through AI technology

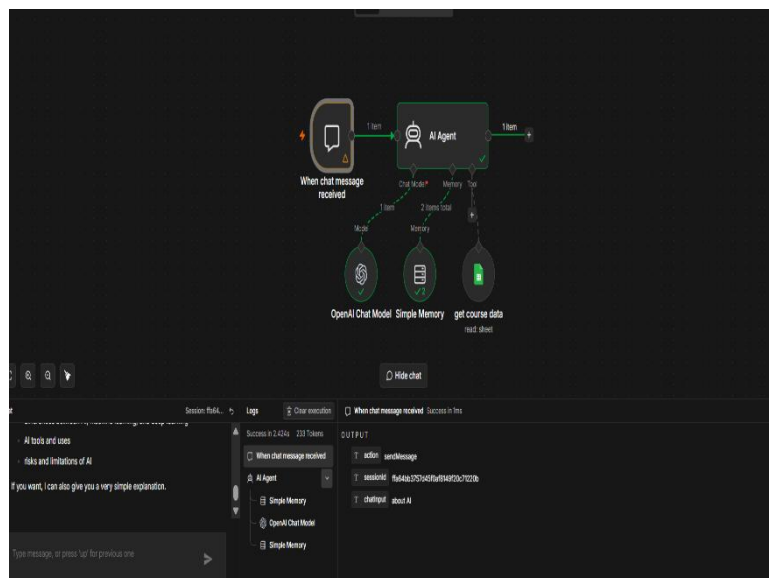
Technologies Used:

- ✓ n8n Workflow Automation
- ✓ OpenAI ChatGPT API
- ✓ Simple Memory Module
- ✓ Google Sheets Integration
- ✓ Cloud-Based Deployment

System

Architecture:

User Query ↓ Chat Trigger ↓ AI Agent (n8n) ↓ OpenAI Chat Model
↓ Memory Management ↓ Google Sheets Data Retrieval ↓ Response
Generation ↓ User



Methodology:

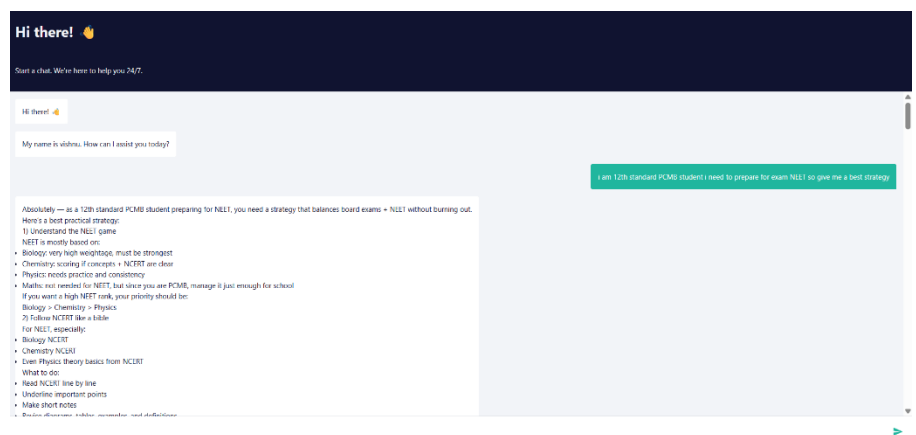
The SmartFlow AI Assistant receives a user's query through a chat interface. The query is processed by the AI Agent workflow created in n8n. The OpenAI Chat Model generates intelligent responses based on the user's input. The memory component stores conversation context, enabling the assistant to remember previous interactions. Google Sheets integration allows retrieval of course-related information when required. The generated response is then delivered back to the user.

Applications:

- ✓ Student Learning Support
- ✓ Academic Query Resolution
- ✓ Self-Learning Assistance
- ✓ Educational Institutions
- ✓ Online Learning Platforms
- ✓ Personal Study Companion

Advantages:

- ✓ Saves Time
- ✓ Easy to Use
- ✓ Available 24/7
- ✓ Personalized Responses
- ✓ Improves Learning Experience
- ✓ Supports Multiple Educational Topics
- ✓ Automated and Scalable



Results:

The SmartFlow AI Assistant successfully answered educational queries and maintained conversational context through memory. The integration of **OpenAI and n8n** enabled efficient response generation and workflow automation. The system demonstrated reliable performance and improved user interaction compared to traditional search-based methods.

Future Scope:

- Voice-Based Interaction
- Multi-Language Support
- Advanced Student Analytics
- Personalized Learning Recommendations
- Integration with Learning Management Systems
- Mobile Application Development

Conclusion:

SmartFlow AI Assistant for Study is an effective AI-based educational support system that combines workflow automation with advanced language models. The project demonstrates how artificial intelligence can improve learning experiences by providing instant, contextual, and intelligent responses.

References:

- ✓ OpenAI Documentation
- ✓ n8n Documentation
- ✓ Artificial Intelligence Fundamentals
- ✓ Workflow Automation Concepts
- ✓ Educational Technology Resources.

